

Chapter 6

1) 1 2 4 8 16 32 64 128

8) The program prints 10 5 3 2 first then prints 1 infinitely

13) for(n = 0; m > 0; n++, m /= 2)

Programming Project 8) file 'calender' is included in zip

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
    unsigned char days, sDay;
```

```
    printf("Enter number of days in month: ");
```

```
    scanf("%u", &days);
```

```
    printf("Enter starting day of the week (1 = Sun , 7 = Sat): ");
```

```
    scanf("%u", &sDay);
```

```
    printf("\n");
```

```
    switch(sDay)
```

```
{
```

```
        case 1: printf(" 1 2 3 4 5 6 7"); break;
```

```

    case 2: printf(" 1 2 3 4 5 6"); break;
    case 3: printf(" 1 2 3 4 5"); break;
    case 4: printf(" 1 2 3 4"); break;
    case 5: printf(" 1 2 3"); break;
    case 6: printf(" 1 2"); break;
    case 7: printf(" 1"); break;
}

```

```

printf("\n");

```

```

for(int i = 9 - sDay; i <= days; i += 7)
{
    for(int j = i; j < i + 7 && j <= days; j++)
    {
        printf("%2u ", j);
    }
    printf("\n");
}

```

```

return 0;
}

```

Chapter 7

1) A) 63

- B) 119
- C) 2748

2) A) Invalid

B) Float

C) Invalid

D) Invalid

E) Float

5) A) Valid

B) Invalid

C) Valid

D) Valid

6) A) Char

B) Short

C) Short

D) Int

9) $i / j + 'a'$ remains as type int